



# The quadratic formula

## HANDOUT

### GOAL

#### In this lesson

**Prove the quadratic formula and use it to solve equations whose discriminant is a perfect square.**

#### BY THE END OF THE LESSON

- ☐ I can prove the quadratic formula by completing the square, explaining why each step is allowed.
- ☐ I can solve a quadratic equation with integer coefficients by the formula when  $D$  is a perfect square, and check a root by substituting it.

1 class period • about 45 min

#### WHAT YOU NEED FROM BEFORE

- ↔ Lesson 1.1 • Quadratic equations: what we can already solve
- ↔ Lesson 1.2 • Solving by completing the square
- ↔ Square roots

### HOOK

#### Two gardens

A rectangular garden is 3 m longer than it is wide. Its area is  $40 \text{ m}^2$ .

Level 1 • threshold

How wide is the garden?

A second garden is also 3 m longer than it is wide, but its area is  $30 \text{ m}^2$ . Now  $x(x + 3) = 30$ , that is,  $x^2 + 3x - 30 = 0$ . No whole number works:  $4 \cdot 7 = 28$  and  $5 \cdot 8 = 40$ .

Completing the square would work, but it means the same four moves again, with new fractions. In lesson 1.2 we made the same four moves every time, only with different numbers. Today we make them **once**, with the letters  $a$ ,  $b$  and  $c$ . The result is a formula that solves any quadratic equation in a few lines.

## The four moves with letters

### The four moves, with numbers

Here are the four moves on lesson 1.2's Example 3. The column on the right explains each step.

|                              |                                     |
|------------------------------|-------------------------------------|
| $4x^2 - 8x + 3 = 0$          | $: 4$                               |
| $x^2 - 2x + \frac{3}{4} = 0$ | $-\frac{3}{4}$                      |
| $x^2 - 2x = -\frac{3}{4}$    | $+ \left(\frac{-2}{2}\right)^2 = 1$ |
| $x^2 - 2x + 1 = \frac{1}{4}$ | left side is a square               |
| $(x - 1)^2 = \frac{1}{4}$    |                                     |

Then  $x - 1 = \frac{1}{2}$  or  $x - 1 = -\frac{1}{2}$ , so  $x = \frac{3}{2}$  or  $x = \frac{1}{2}$ .

From one equation to the next, only the numbers 4,  $-8$  and 3 change. Now put the letters  $a$ ,  $b$  and  $c$  in their place.

### The same moves with a, b and c

Take any quadratic equation  $ax^2 + bx + c = 0$ . Make the same moves:

|                                                                          |                                 |
|--------------------------------------------------------------------------|---------------------------------|
| $ax^2 + bx + c = 0$                                                      | $: a$                           |
| $x^2 + \frac{b}{a}x + \frac{c}{a} = 0$                                   | $-\frac{c}{a}$                  |
| $x^2 + \frac{b}{a}x = -\frac{c}{a}$                                      | $+ \left(\frac{b}{2a}\right)^2$ |
| $x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a}$ | left side is a square           |
| $\left(x + \frac{b}{2a}\right)^2 = \frac{b^2}{4a^2} - \frac{c}{a}$       |                                 |

Why each step is allowed:

1. **Divide by  $a$ .** A quadratic equation has  $a \neq 0$ , so we may divide both sides by  $a$ . The roots do not change.
2. **Move the constant term.** Subtract  $\frac{c}{a}$  from both sides.
3. **Add the square of half the coefficient of  $x$ .** The coefficient of  $x$  is now  $\frac{b}{a}$ . Half of it is  $\frac{b}{2a}$ , and its square is  $\frac{b^2}{4a^2}$ . Add it to **both** sides.
4. **Write the left side as a square.** This is lesson 1.2's rule, with  $\frac{b}{a}$  in place of  $b$ .

## One fraction on the right

Bring the right side to the common denominator  $4a^2$ :

$$\frac{b^2}{4a^2} - \frac{c}{a} = \frac{b^2}{4a^2} - \frac{4ac}{4a^2} = \frac{b^2 - 4ac}{4a^2}.$$

So **every** quadratic equation can be written as

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}.$$

The number  $b^2 - 4ac$  decides what happens next. It is called the **discriminant** and is written  $D$ :

$$D = b^2 - 4ac.$$

The denominator  $4a^2$  is positive, because  $a \neq 0$ . So the right side,  $\frac{D}{4a^2}$ , has the same sign as  $D$ .

## Take the root

Now we have  $(x + p)^2 = d$  from lesson 1.1, with  $p = \frac{b}{2a}$  and  $d = \frac{D}{4a^2}$ .

If  $D \geq 0$ , the bracket equals  $+\sqrt{D/(4a^2)}$  or  $-\sqrt{D/(4a^2)}$  (when  $D = 0$ , both are 0):

$$x + \frac{b}{2a} = \pm \sqrt{\frac{D}{4a^2}} = \pm \frac{\sqrt{D}}{2|a|}.$$

(The root of  $4a^2$  is  $2|a|$ , because a root is never negative.) If  $a < 0$ , then  $2|a| = -2a$ . But the sign  $\pm$  already contains both signs, so  $\pm\sqrt{D}/(2|a|)$  and  $\pm\sqrt{D}/(2a)$  are the same two numbers. Subtract  $\frac{b}{2a}$  from both sides:

$$x = -\frac{b}{2a} \pm \frac{\sqrt{D}}{2a} = \frac{-b \pm \sqrt{D}}{2a}.$$

If  $D < 0$ , the right side is negative. No square is negative, so the equation has **no real roots**. Lesson 1.5 looks at  $D < 0$  and  $D = 0$  closely.

## The quadratic formula

The roots of  $ax^2 + bx + c = 0$  ( $a \neq 0$ ) are

$$x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$$

where  $D = b^2 - 4ac$ ; the formula is used when  $D \geq 0$ . Written out, as on the PUPP formula sheet:

$$x_1 = \frac{-b - \sqrt{D}}{2a},$$
$$x_2 = \frac{-b + \sqrt{D}}{2a}.$$

### How to use it:

1. Write the equation as  $ax^2 + bx + c = 0$  and read  $a, b, c$  **with their signs**.
2. Work out  $D = b^2 - 4ac$ . Put negative numbers in brackets:  $(-3)^2 = 9$ .
3. Work out  $\sqrt{D}$ . In this lesson  $D$  is always a perfect square, so  $\sqrt{D}$  is a whole number.
4. Write  $x_1$  and  $x_2$ . The **whole** numerator  $-b \pm \sqrt{D}$  is divided by  $2a$ .
5. Check a root by substituting it into the equation.

### WORKED EXAMPLES

#### Worked examples

##### Example 1 • The first garden: $x^2 + 3x - 40 = 0$

###### STANDARD FORM

$x(x + 3) = 40$  gives  $x^2 + 3x - 40 = 0$ :  $a = 1, b = 3, c = -40$ .

###### DISCRIMINANT

$D = 3^2 - 4 \cdot 1 \cdot (-40) = 9 + 160 = 169$ , and  $\sqrt{169} = 13$ .

###### FORMULA

$$x_1 = \frac{-3 - 13}{2 \cdot 1} = -8, \quad x_2 = \frac{-3 + 13}{2 \cdot 1} = 5.$$

###### ANSWER

A width cannot be negative, so the garden is 5 m wide and 8 m long, as we found by trying. Check:  $5 \cdot 8 = 40$  ✓.

##### Example 2 • When $a \neq 1$ : $2x^2 - 3x - 2 = 0$

###### COEFFICIENTS

$a = 2, b = -3, c = -2$ . So  $-b = 3$ .

###### DISCRIMINANT

$D = (-3)^2 - 4 \cdot 2 \cdot (-2) = 9 + 16 = 25$ , and  $\sqrt{25} = 5$ .

###### FORMULA

$$x_1 = \frac{3 - 5}{2 \cdot 2} = -\frac{1}{2}, \quad x_2 = \frac{3 + 5}{2 \cdot 2} = 2.$$

###### CHECK

$2 \cdot 2^2 - 3 \cdot 2 - 2 = 8 - 6 - 2 = 0$  ✓.

## SUMMARY

### What to remember

- The **discriminant** of  $ax^2 + bx + c = 0$  is  $D = b^2 - 4ac$ .
- The **quadratic formula**: if  $D \geq 0$ , the roots are  $x_{1,2} = \frac{-b \pm \sqrt{D}}{2a}$  ( $x_1$  with the minus,  $x_2$  with the plus, as on the formula sheet).
- It comes from the four moves of completing the square, made once with  $a$ ,  $b$  and  $c$ . We may divide by  $a$  because  $a \neq 0$ .
- **How to use it**: write  $ax^2 + bx + c = 0$ ; read  $a$ ,  $b$ ,  $c$  with their signs; work out  $D$  with brackets around negative numbers; divide the **whole** numerator by  $2a$ ; check a root by substituting it.
- In a word problem, keep only the roots that make sense.

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## NEXT LESSON

**The second garden,  $x^2 + 3x - 30 = 0$ , has  $D = 129$ , which is not a perfect square. How do we write its width exactly, and how wide is it roughly?**

[Lesson 1.4 · The formula with irrational roots →](#)